

Test Report - VxMark Vibration Test

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Summary

The VxMark ballot-marking device (BMD) showed no change in function after standard 3-axis vibration testing simulating truck transport with an external test vendor.

Applicable VVSG Requirements

- 2.7-E – Ability to support transport and storage physical environment conditions – non-operating
 - Test Assertions, 2.7-E: Withstand *MIL-STD-810H, Method 514.8, Procedure I – General Vibration, Transportation*.

Devices Under Test

- VxMark, v4.1, image v4.0.4

Purpose

VxMark must resist damage due to expected transport. This was tested using standard methods detailed in MIL-STD-810H, Method 514.8, Procedure I – General Vibration, Transportation, and previously used by Element in Longmont, CO for other VxSuite products. A packaged VxMark was planned to undergo 1 hour of vibration along each of its 3 axes, totaling 3 hours, with vibration profiles matching that defined in the test standard, with the test vendor Austin Reliability Labs in Austin, TX. The VxMark system printer and UPS were not tested, because these are two commercial-off-the-shelf products with histories of shipping successfully in their packaging.

Materials

- VxMark unit (MK-14-018)
 - Imaged with v4.0.4 production app
 - Configured with summary ballot test election
- VxMark printer (HP4001dn, VNL0728678) and paper
- Goldenmate 1500VA/1000W UPS (JUO2HE390707)

- Electrodynamic shaker, single axis (Austin Reliability Labs)

Procedures

1. Check the VxMark BMD for normal function before vibration tests.
 - a. Follow tests of normal function (*VxMark Tests of Normal Function*) to confirm the test unit functions normally before any disruptive tests.
 - b. Document tests of normal function. Record any pre-existing problems. Use the pre-existing template.
2. Package the VxMark according to standard internal protocols, including protective internal padding and standard tape seal.
3. Test the packaged unit for vibration, sequentially in vertical, transverse, and longitudinal orientations. Vibrate the unit according to *MIL-STD-810H, Method 514.8, Procedure I – General Vibration, Transportation*, or with the profile summarized in [Appendix A](#).
4. Unpackage the VxMark unit.
5. Check the VxMark unit for normal function after vibration tests.

Results

Normal function of the VxMark BMD was confirmed before testing, as documented in [Appendix B](#).

After packaging according to standard VotingWorks protocols, the VxMark underwent standard vibration testing. Photos of the test package are shown in [Appendix C](#). Actual vibration profiles from the control system and the on the unit are shown in [Appendix D](#).

After vibration testing, the unit was unpackaged and tested for normal function again, showing no changes or loss in function. This is documented in [Appendix E](#).

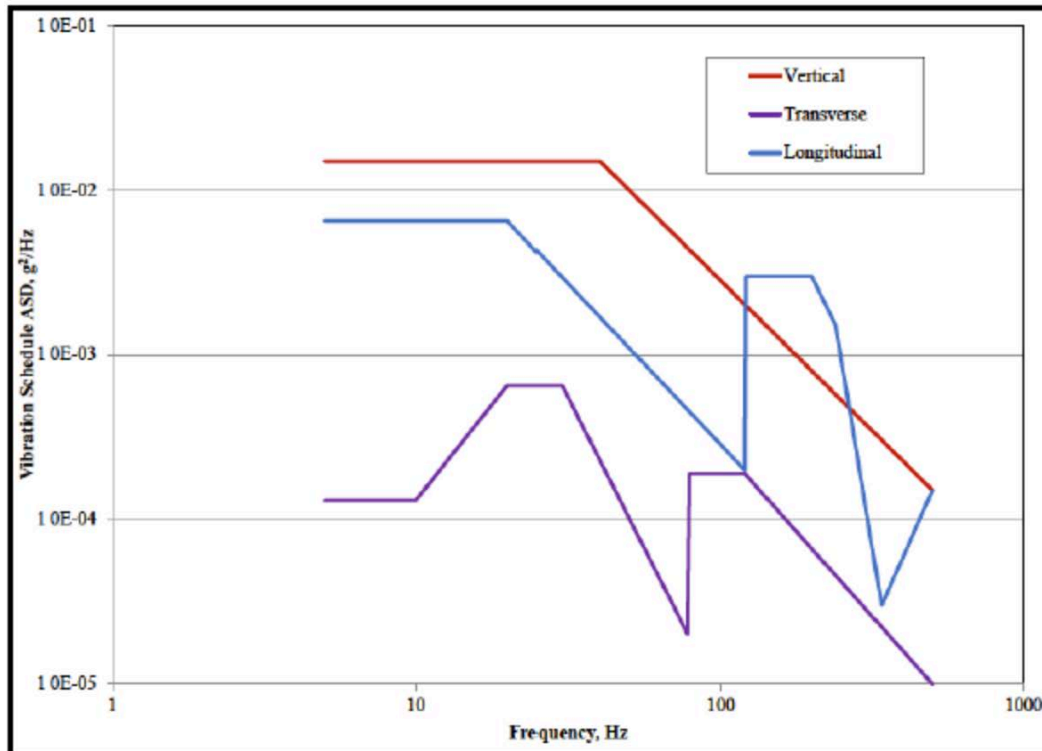
Additional bench handling drop tests were performed on the unpackaged unit to further stress test its hardware, following similar procedures as in a more comprehensive previous test documented in *Test Report - VxMark Drop Test, Bench Handling*. No loss in function was found, documented in [Appendix F](#).

Conclusions

- The VxMark BMD is likely to pass standard vibration tests required by VVSG 2.7-E.
- No loss in function or loose parts were found after standard 3-axis vibration testing, 1 hour each axis, followed by bench handling drop tests.

Appendix A: Vibration Profile Standard

Profiles taken below from *MIL-STD-810H, Method 514.8, Procedure I – General Vibration, Transportation*.



Vertical		Transverse		Longitudinal	
Frequency, Hz	ASD, g²/Hz	Frequency, Hz	ASD, g²/Hz	Frequency, Hz	ASD, g²/Hz
5	0.015	5	0.00013	5	0.0065
40	0.015	10	0.00013	20	0.0065
500	0.00015	20	0.00065	120	0.0002
		30	0.00065	121	0.003
		78	0.00002	200	0.003
		79	0.00019	240	0.0015
		120	0.00019	340	0.00003
		500	0.00001	500	0.00015
rms = 1.08 g		rms = 0.21 g		rms = 0.76 g	

Appendix B: VxMark Normal Function Checklist - 20260511 - Before vibration

Corresponds to detailed doc here:		<i>VxMark Tests of Normal Function</i>	
	Person(s) testing & location:	Pius Wong	VxAustin
	Date & time tested	5/11/2026	4:00 PM
	System tested & Unit/Model No.:	MK-14-018	
	Before what other test?	Vibration testing	
	After what other test?	Continuous operations	
Always do these:			
No.	Test	Pass / Fail / Mixed?	Notes
1	Marks and prints valid ballots.	Pass	
2	Visually displays instructions and accurate feedback to users.	Pass	
3	Audibly plays instructions and accurate feedback to users.	Pass	
4	Accepts user inputs on the touchscreen.	Pass	
5	Accepts user inputs on the ATI and PAT.	Pass	
6	Accepts inputs on the barcode scanner.	Pass	
Sometimes do these (after environmental, vibrational, or other mechanical tests):			
No.	Test	Pass / Fail / Mixed?	Notes
7	Maintains physical security, restricting administrative access and preventing tampering without evidence.	Pass	

8	Protects internal electronics and subsystems.	Pass	
Inspection of parts, after tests of normal function above:			
No.	Part	Visual condition ?	notes
1	Computer	n/a	not opened up
2	Touchscreen	Pass	
3	Power Supply, internal	n/a	not opened up
4	Smart Card Reader	Pass	
5	USB-A receptacle	Pass	
6	USB 3 Switch	n/a	not opened up
7	Barcode Scanner	Pass	
8	Cable Assemblies	Pass	
9	Uninterruptible Power Supply (external to unit)	Pass	
10	Printer (external to unit)	Pass	
11	Outer case	Pass	
12	Inner panels around electronics	Pass	
13	USB cover	Pass	
14	Security tie points	Pass	
15	Fasteners, both external and internal	Pass	
16	Wires, ports, and electrical connections	n/a	not opened up
18	other....		

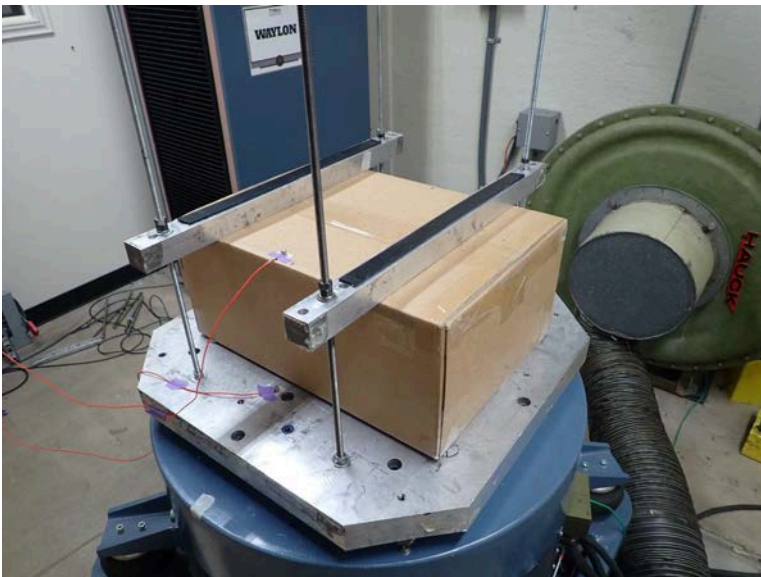
Appendix C: Photos of Testing

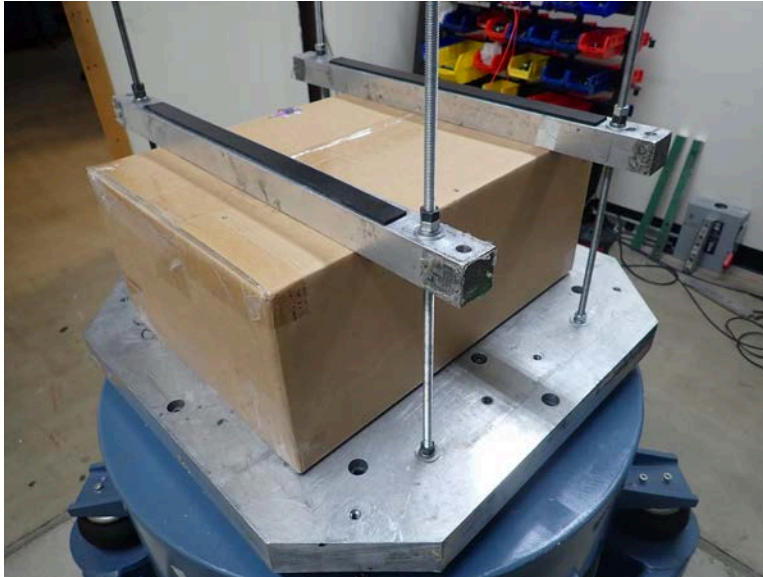
1. Pre-Test



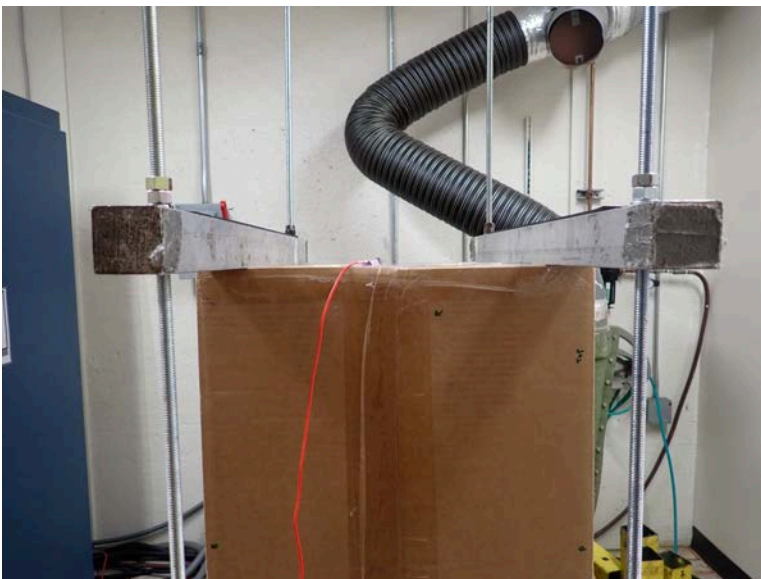
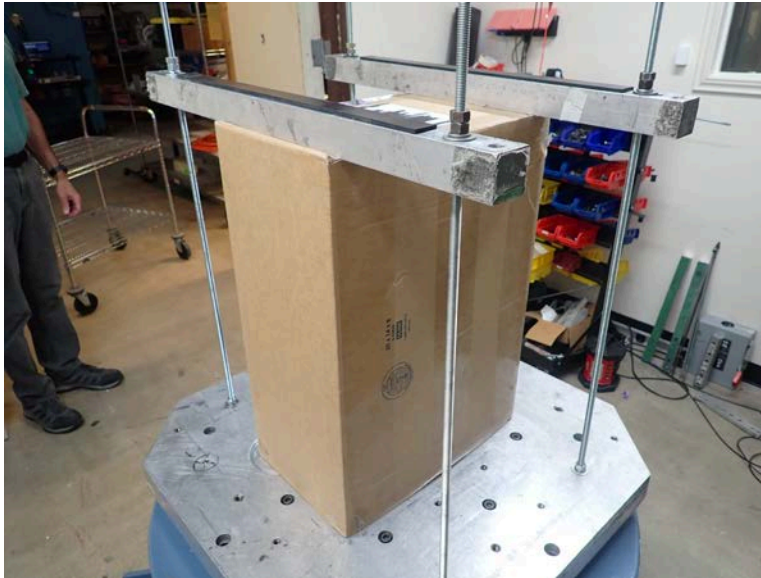


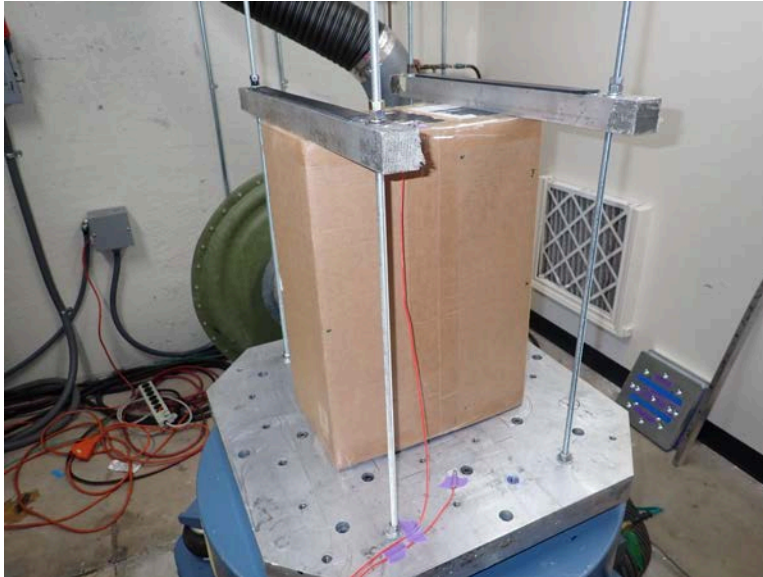
2. Post-Vertical





3. Post-Longitudinal





4. Post-Transverse / Post-All Testing

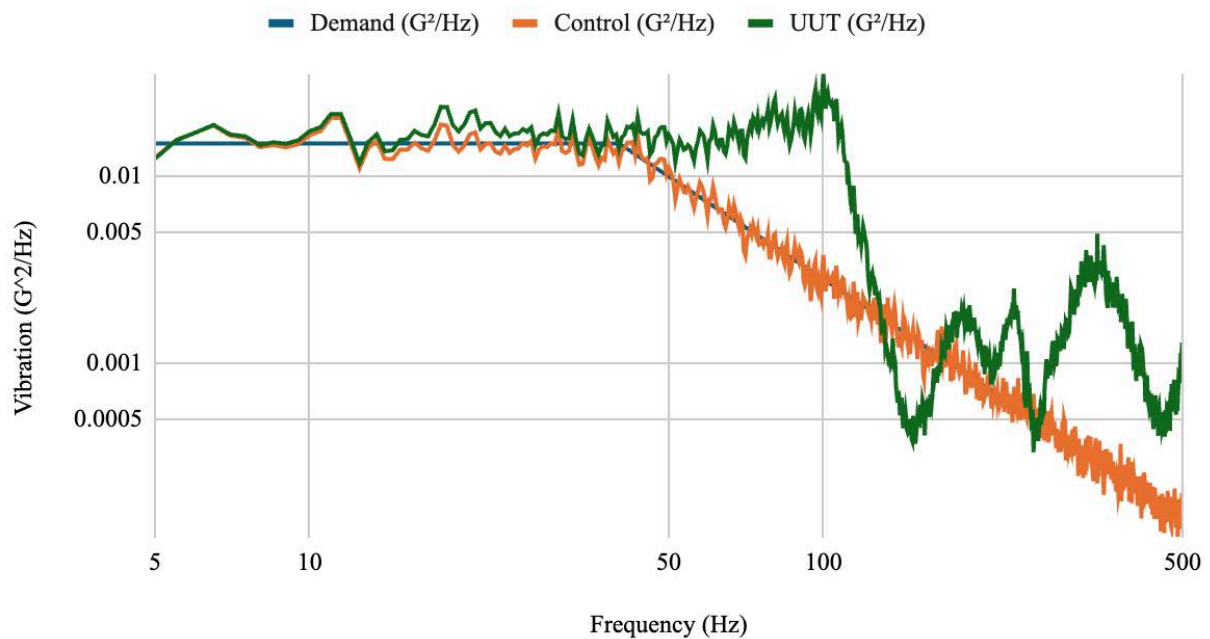




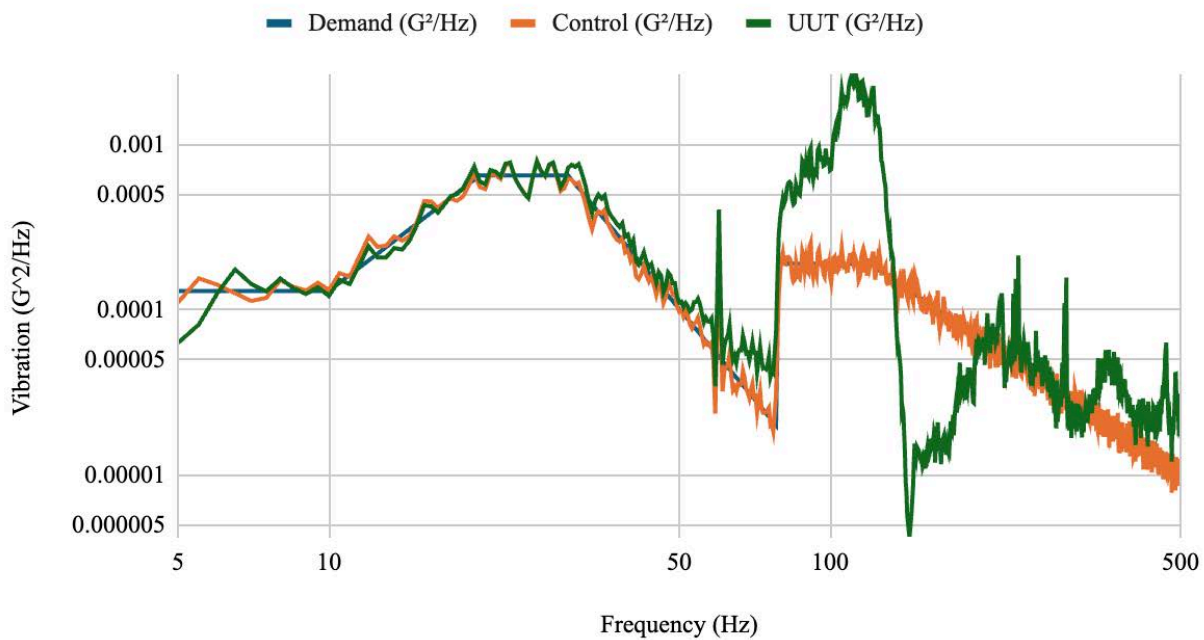
Appendix D: Vibration Profiles

In the graphs below, control vibration is shown in yellow, and vibration of the unit under test (UUT) is shown in green.

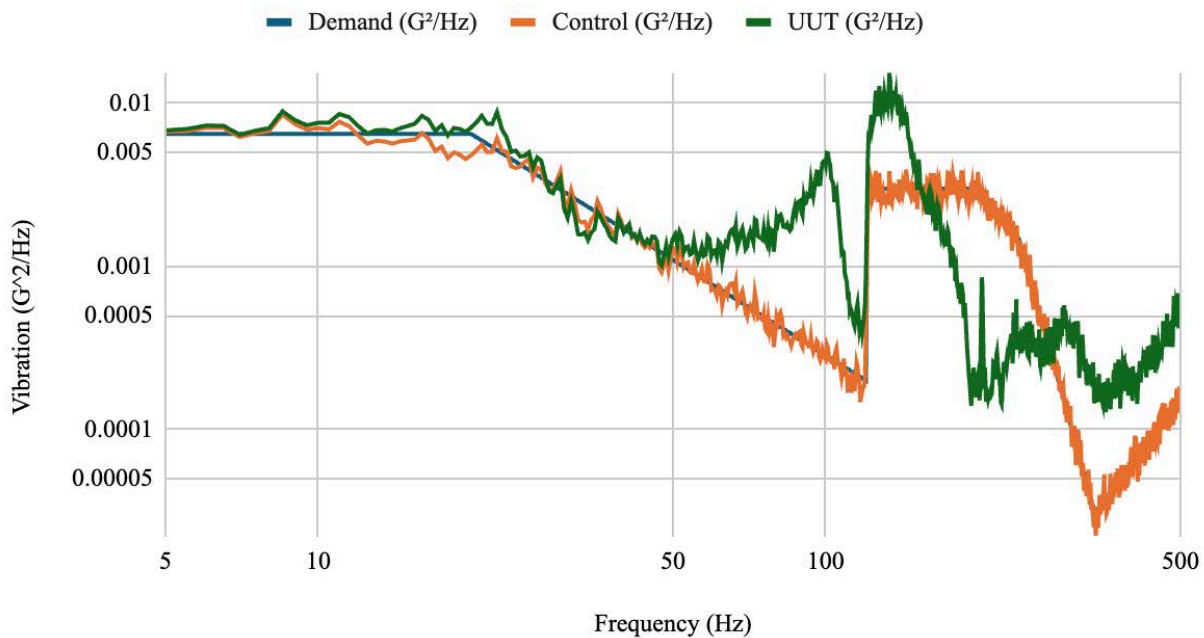
VxMark vibration profile, Vertical axis, 5/26



VxMark vibration profile, Transverse axis, 5/26



VxMark vibration profile, Longitudinal axis, 5/26



Appendix E: VxMark Normal Function Checklist - 20260512 - After vibration

Corresponds to detailed doc here:		<i>VxMark Tests of Normal Function</i>	
	Person(s) testing & location:	Pius Wong	VxAustin
	Date & time tested	5/12/2026	5:17 PM
	System tested & Unit/Model No.:	MK-14-018	
	Before what other test?	post-vibration drop testing	
	After what other test?	Vibration testing	
Always do these:			
No.	Test	Pass / Fail / Mixed?	Notes
1	Marks and prints valid ballots.	Pass	
2	Visually displays instructions and accurate feedback to users.	Pass	
3	Audibly plays instructions and accurate feedback to users.	Pass	
4	Accepts user inputs on the touchscreen.	Pass	
5	Accepts user inputs on the ATI and PAT.	Pass	
6	Accepts inputs on the barcode scanner.	Pass	
Sometimes do these (after environmental, vibrational, or other mechanical tests):			
No.	Test	Pass / Fail / Mixed?	Notes
7	Maintains physical security, restricting administrative access and preventing tampering without evidence.	Pass	

8	Protects internal electronics and subsystems.	Pass	
Inspection of parts, after tests of normal function above:			
No.	Part	Visual condition ?	notes
1	Computer	n/a	not opened up
2	Touchscreen	Pass	
3	Power Supply, internal	n/a	not opened up
4	Smart Card Reader	Pass	
5	USB-A receptacle	Pass	
6	USB 3 Switch	n/a	not opened up
7	Barcode Scanner	Pass	
8	Cable Assemblies	Pass	
9	Uninterruptible Power Supply (external to unit)	Pass	
10	Printer (external to unit)	Pass	
11	Outer case	Pass	
12	Inner panels around electronics	Pass	
13	USB cover	Pass	
14	Security tie points	Pass	
15	Fasteners, both external and internal	Pass	
16	Wires, ports, and electrical connections	n/a	not opened up
18	other....		

Appendix F: VxMark Normal Function Checklist - 20260514 - After drop test

Corresponds to detailed doc here:		<i>VxMark Tests of Normal Function</i>	
	Person(s) testing & location:	Pius Wong	VxAustin
	Date & time tested	5/14/2026	4:00 PM
	System tested & Unit/Model No.:	MK-14-018	
	Before what other test?	n/a	
	After what other test?	post-vibration drop testing	
Always do these:			
No.	Test	Pass / Fail / Mixed?	Notes
1	Marks and prints valid ballots.	Pass	
2	Visually displays instructions and accurate feedback to users.	Pass	
3	Audibly plays instructions and accurate feedback to users.	Pass	
4	Accepts user inputs on the touchscreen.	Pass	
5	Accepts user inputs on the ATI and PAT.	Pass	
6	Accepts inputs on the barcode scanner.	Pass	
Sometimes do these (after environmental, vibrational, or other mechanical tests):			
No.	Test	Pass / Fail / Mixed?	Notes
7	Maintains physical security, restricting administrative access and preventing tampering without evidence.	Pass	

8	Protects internal electronics and subsystems.	Pass	
Inspection of parts, after tests of normal function above:			
No.	Part	Visual condition ?	notes
1	Computer	n/a	not opened up
2	Touchscreen	Pass	
3	Power Supply, internal	n/a	not opened up
4	Smart Card Reader	Pass	
5	USB-A receptacle	Pass	
6	USB 3 Switch	n/a	not opened up
7	Barcode Scanner	Pass	
8	Cable Assemblies	Pass	
9	Uninterruptible Power Supply (external to unit)	Pass	
10	Printer (external to unit)	Pass	
11	Outer case	Pass	
12	Inner panels around electronics	Pass	
13	USB cover	Pass	
14	Security tie points	Pass	
15	Fasteners, both external and internal	Pass	
16	Wires, ports, and electrical connections	n/a	not opened up
18	other....		headphones door popped open once, no damage, could be closed